

Alpha value of 0.01 was used for the entirety of this homework to allow for slow descent over an abundant amount of iterations, 2000 to be exact. This is to ensure there are enough chances for the minimum to be found while preventing overshooting.

Problem 1

1. Develop a gradient descent training and evaluation code..
 - a. With the 5 unscaled input features, gradient descent diverged due to the large scale of price and area. Numerical Overflow occurred.
 - b. With the 11 unscaled input features, gradient descent diverged due to the large scale of price and area. Numerical Overflow occurred even with the increased feature count, pointing to a scale issue as discussed in part a.
2. Repeat problem 1 with feature scaling.
 - a. Normalization final theta = [2748676.78, 2241568.18, 1403394.61, 1890740.47, 1700157.82, 1465097.77]
Final training cost: 8.87×10^{11}
Final validation cost: 7.87×10^{11}

Standardization final theta = [4795729.21, 729932.49, 79337.83, 641263.74, 463853.70, 287330.71]
Final training cost: 7.61×10^{11}
Final validation cost: 7.98×10^{11}

Standardization achieved lower training cost, but normalization achieved lower validation cost. Since validation performance better reflects the model performance, normalization may be the preferable choice here, even though standardization fits the training data more closely.

- b. Normalization - final training cost: 6.59×10^{11} , final validation cost: 5.88×10^{11}
Standardization - final training cost: 5.88×10^{11} , final validation cost: 6.25×10^{11}

Same results as 2a, standardization wins on training cost, normalization wins on validation cost. Both scaling methods also produced much lower cost than the 5-variable models in 2a, since the added variables give the model more information to explain price.

3. Apply regularization
 - a. Standardization was used for the regularization because of its lower training cost in problem 2. Values of 0.01, 1, 5, and 10 were tested for lambda.

Final training cost: 7.61×10^{11}

Final validation cost: 7.95×10^{11} (at $\lambda=0.01$)

Compared to Problem 2a's standardization baseline (validation cost 7.98×10^{11}), $\lambda=0.01$ gave a small improvement in validation cost with minimal change in training cost. $\lambda=0.01$ was the lowest value tested and produced the best result among those tried.

- b. Same approach applied to the 11-variable model. $\lambda=0.01$ was used, compared against $\lambda=0.03$.

Final training cost: 5.88×10^{11}

Final validation cost: 6.24×10^{11}

Regularization produced only a small improvement over Problem 2b's standardization baseline. $\lambda=0.01$ gave the lowest cost out of both the values tested.